INMO(36^{th}) – 2021

Time – 4 Hrs

Instructions:

- **Calculators (in any form) and protractors are not allowed.**
- **Ruler and compass allowed.**
- **Answer all the questions.**
- **All Questions carry equal marks. Maximum marks: 102.**
- **Answer to each question should start on a new page. Clearly indicate the question number.**

1. Suppose $r \geq 2$ is an integer, and let $m_1, n_1, m_2, n_2, ..., m_r, n_r$ be 2r integers such that $|m_i n_j - m_j n_i| = 1$ for any two integers I and j satisfying $1 \leq i < j \leq r$. Determine the maximum possible values of r.
2. Find all pairs of integers (a, b) such that each of the two cubic polynomials $x^3 + bx + a$ and $x^3 + ax + b$ has all the roots to be integer
3. Betal marks 2021 points on the plane such that no three are collinear, and draws all possible line segments joining these. He then chooses any 1011 of these lines segments, and marks their midpoints. Finally he chooses a line segment whose midpoint is not marked yet, and challenges Vikram to construct its midpoints using only a straightedge. Can Vikram always complete the challenge?
   Note: A straightedge is an infinitely long ruler without markings, which can only be used to draw the line joining any two given distinct points.
4. A magician and a detective play a game. The magician lays down cards numbered from 1 to 52 face-down on a table. On each move, the detective can point to two cards and inquire if the numbers on them are consecutive. The magician replies truthfully. After a finite number of moves the detective points to two cards. She wins if the numbers on these cards are consecutive, and loses otherwise. Prove that the detective can guarantee a win if and only if she allowed to ask at least 50 questions.
5. In a convex quadrilateral ABCD, $\angle ABD = 30^0, \angle BCA = 75^0, \angle ACD = 75^0$ and CD = CB. Extend CB to meet the circumcircle of triangle DAC at E. Prove that CE = BD.
6. Let $\Re[x]$ be the set of all polynomials with real coefficients, and let deg P denote the degree of the polynomial P. Find all functions $f : \Re[x] \to \Re[x]$ satisfying the following conditions:
   - F maps the zero polynomial to itself.
   - For any non-zero polynomial $P \in \Re[x]$, $\deg f(P) \leq 1 + \deg P$
   - For any two polynomials $P, Q \in \Re[x]$, the polynomials $P - f(Q)$ and $Q - f(P)$ have the same set of real roots.

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